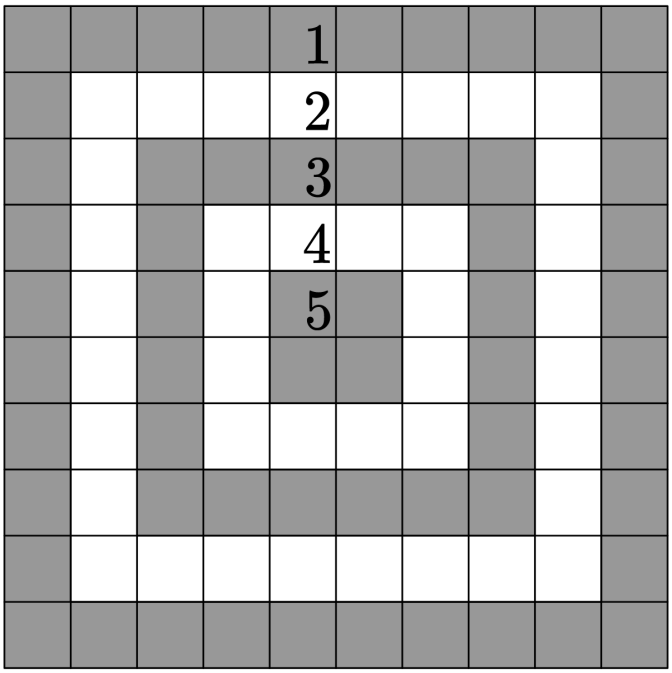
C. Target Practice

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: implementation, math | https://codeforces.com/problemset/problem/1873/C

A 10 × 10 target is made out of five "rings" as shown. Each ring has a different point value: the outermost ring — 1 point, the next ring — 2 points, ..., the center ring — 5 points.



Vlad fired several arrows at the target. Help him determine how many points he got.

Input

The input consists of multiple test cases. The first line of the input contains a single integer t ($1 \le t \le 1000$) — the number of test cases.

Each test case consists of 10 lines, each containing 10 characters. Each character in the grid is either X (representing an arrow) or . (representing no arrow).

Output

For each test case, output a single integer — the total number of points of the arrows.

Example

Input

```
4
X.........
..........
.......X..
.....X....
......X...
..........
.........X
..X.......
..........
.........X
..........
..........
..........
..........
..........
..........
..........
..........
..........
..........
..........
..........
..........
....X.....
..........
..........
..........
..........
..........
XXXXXXXXXX
XXXXXXXXXX
XXXXXXXXXX
XXXXXXXXXX
XXXXXXXXXX
XXXXXXXXXX
XXXXXXXXXX
XXXXXXXXXX
XXXXXXXXXX
XXXXXXXXXX
```

Output

```
17
0
5
220
```

Note

In the first test case, there are three arrows on the outer ring worth 1 point each, two arrows on the ring worth 3 points each, and two arrows on the ring worth 4 points each. The total score is $3 \times 1 + 2 \times 3 + 2 \times 4 = 17$.

In the second test case, there aren't any arrows, so the score is 0.